

This is explained as follows:

Suppose that X represents the total order shares, Y represents the shares that have traded, and $(X - Y)$ represents the shares that have not yet traded (unexecuted shares). Then,

$$\theta = \frac{Y}{X} \text{ represents the percentage of shares traded.}$$

$$(1 - \theta) = \frac{(X - Y)}{X} \text{ represents the percentage of shares not yet traded.}$$

Additionally, let,

$E_0[\cdot]$ = initial time expectation. The expected cost and prices at the beginning of trading.

$E_t[\cdot]$ = time expectation. The expected cost and prices at the current point in time.

$E_0[Cost]$ = C^* initial estimated cost.

P_t = market price at time t .

$\bar{P}_t$ = average execution price of the Y shares at time t .

Then the cost components are:

Realized Cost: the actual cost of the Y traded shares. This is:

$$Realized(Cost(t)) = \theta \cdot (\bar{P}_t - P_0) \cdot Side \quad (8.27)$$

Momentum Cost: the price movement in the stock from the time trading began to the current time. This price movement results in either a sunk cost or savings to the investor. Momentum cost is applied to the number of unexecuted shares since these are the shares that will realize the cost or realize the savings. This is:

$$E_t[Momentum(Cost)] = (1 - \theta) \cdot (P_t - P_0) \cdot Side \quad (8.28)$$

Remaining Market Impact: the remaining market impact cost of the unexecuted shares. This is the expected price impact that will result from trading the unexecuted shares in the current market environment (liquidity and volatility) and with the current trading rate. Mathematically this is:

$$E_t[MI'] = (1 - \theta) \cdot (b_1 \cdot I' \cdot \alpha_t + (1 - b_1) \cdot I') \quad (8.29)$$

Alpha Trend: the cost that will result due to the alpha trend over the trading horizon. Mathematically this is:

$$E_t[Alpha'] = (1 - \theta) \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha_t} \cdot \mu_t \quad (8.30)$$

where μ_t is the alpha trend over the trading horizon expressed in \$/share.